

Linear regression Session – 2

Linear regression equation: $y = b_0 + b_1 * x_1 + b_2 * x_2 + \dots + b_n * x_n$

- Generally we have input column and we also have target data
- input columns we denoted with $x_1, x_2, \dots, x_n$
- output column or target column denoted with y
- this target column y , we called actual output : y_{actual}
- we have one more name : Ground truth data
- model will train by passing input data and actual output data
- Once model is developed we will pass again input data only
- so model also will give some outputs, this output is called as predictions : $y_{predictions}$
- in order to evaluate the model performance we need to compare y_{actual} with $y_{prediction}$

$$Sales = b_0 + b_1 * SM + b_2 * TV + b_3 * NP$$

$$\text{assume } b_0 = 1, b_1 = 5, b_2 = 2, b_3 = 2$$

$$Sales = 1 + 5 * SM + 2 * TV + 2 * NP$$

SM	TV	NP	Sales = y_{actual}
300	400	500	1000
200	150	500	2000
100	200	300	500
400	500	200	3000
300	150	150	1000

<i>SM</i>	<i>TV</i>	<i>NP</i>	<i>Sales</i> <i>= y_actual</i>	<i>sales</i> <i>= y_predictions</i> <i>= 1 + 5 * SM</i> <i>+ 2 * TV + 2</i> <i>* NP</i>	Error: <i>y_actual -</i> <i>y_predictions</i>
300	400	500	1000	3301	Error1=-2301
200	150	500	2000	2301	Error2=-301
100	200	300	500	1501	Error3=-1001
400	500	200	3000	3401	Error4=-401
300	150	150	1000	2101	Error5=-1101

$$1 + 5 * 300 + 2 * 400 + 2 * 500$$

$$1 + 5 * 200 + 2 * 150 + 2 * 500$$

$$1 + 5 * 100 + 2 * 200 + 2 * 300$$

$$1 + 5 * 400 + 2 * 500 + 2 * 200$$

$$1 + 5 * 300 + 2 * 150 + 2 * 150$$

Error:

$$Error = (y_{actual} - y_{prediction}) \text{ or } (y_a - y_p)$$

- for every observations has actual output,
- for every observation model will give prediction
- will compare y_{actual} and $y_{prediction}$ observation by observation

$$Error_1 = (y_{a_1} - y_{p_1})$$

$$Error_2 = (y_{a_2} - y_{p_2})$$

$$Error_3 = (y_{a_3} - y_{p_3})$$

$$Error_4 = (y_{a_4} - y_{p_4})$$

$$Error_5 = (y_{a_5} - y_{p_5})$$

Total error

$$TE = e_1 + e_2 + e_3 + e_4 + e_5$$

$$TE = (y_{a_1} - y_{p_1}) + (y_{a_2} - y_{p_2}) + (y_{a_3} - y_{p_3}) + (y_{a_4} - y_{p_4}) + (y_{a_5} - y_{p_5})$$

$$TE = \sum_{i=1}^5 (y_{a_i} - y_{p_i})$$

$$TE = e_1 + e_2 + e_3 \dots e_n$$

$$TE = (y_{a_1} - y_{p_1}) + (y_{a_2} - y_{p_2}) + (y_{a_3} - y_{p_3}) + \dots + (y_{a_n} - y_{p_n})$$

$$TE = \sum_{i=1}^n (y_{a_i} - y_{p_i})$$

Problem of Total Error:

- *Total error is a summation of all individual errors*
- *one error might be positive and another error might be negative*
- *when we do sum of some positive values and some negative values*
- *there might be a chance the total error becomes zero*
- *we are seeing individual errors, but total error zero where math fails*
- *this is same analogy of statistics mean deviation part*

In order to avoid we need to do square of the errors:

SUM OF SQUARE ERRORS (SSE)

ERRORS:

$$Error_1 = (y_{a_1} - y_{p_1})$$

$$Error_2 = (y_{a_2} - y_{p_2})$$

$$Error_3 = (y_{a_3} - y_{p_3})$$

$$Error_4 = (y_{a_4} - y_{p_4})$$

$$Error_5 = (y_{a_5} - y_{p_5})$$

SQUARE ERRORS

$$(Error_1)^2 = (y_{a_1} - y_{p_1})^2$$

$$(Error_2)^2 = (y_{a_2} - y_{p_2})^2$$

$$(Error_3)^2 = (y_{a_3} - y_{p_3})^2$$

$$(Error_4)^2 = (y_{a_4} - y_{p_4})^2$$

$$(Error_5)^2 = (y_{a_5} - y_{p_5})^2$$

$$SSE = e_1^2 + e_2^2 + e_3^2 + e_4^2 + e_5^2$$

$$SSE = (y_{a_1} - y_{p_1})^2 + (y_{a_2} - y_{p_2})^2 + (y_{a_3} - y_{p_3})^2 + (y_{a_4} - y_{p_4})^2 + (y_{a_5} - y_{p_5})^2$$

$$SSE = \sum_{i=1}^5 (y_{a_i} - y_{p_i})^2$$

$$SSE = \sum_{i=1}^n (y_{a_i} - y_{p_i})^2$$

MEAN SQUARE ERRORS (MSE)

$$MSE = \frac{1}{n} * SSE$$

$$MSE = \frac{1}{n} * \sum_{i=1}^n (y_{a_i} - y_{p_i})^2$$

ROOT MEAN SQUARE ERRORS (RMSE)

$$RMSE = \sqrt{MSE}$$

$$RMSE = \sqrt{\frac{1}{n} * \sum_{i=1}^n (y_{a_i} - y_{p_i})^2}$$

$$Error = (y_a - y_p)$$

$$error^2 = (y_a - y_p)^2$$

$$total\ error = \sum_{i=1}^n (y_{a_i} - y_{p_i})$$

$$SSE = \sum_{i=1}^n (y_{a_i} - y_{p_i})^2$$

$$MSE = \frac{1}{n} * \sum_{i=1}^n (y_{a_i} - y_{p_i})^2$$

$$RMSE = \sqrt{\frac{1}{n} * \sum_{i=1}^n (y_{a_i} - y_{p_i})^2}$$

- Error also called as Residual
 - SSE also called as Residual sum of Squares(RSS)
- Single error also called as Loss function
- Sum of square errors also called as : Residual sum of squares (RSS)
- Instead of e_1^2, e_2^2 you might have see r_1^2, r_2^2
- MSE also called as Cost function
- Cost function means sum of all errors (Losses)
- Cost function denoted with : J

$$cost\ function = J = \frac{1}{n} * \sum_{i=1}^n (y_{a_i} - y_{p_i})^2$$

Goal: we need to Find suitable coefficients, in order to minimize the cost function

Suppose $f(x) = y$ means, the equation y has only x variables

$$y = x + 2$$

$$f(x) = x + 2$$

$$f(x) = y$$

similarly

$$y = 2x + 3z$$

$$f(x, z) = 2x + 3z$$

$$f(x, z) = y$$

$f(x, z)$ means equation has combination of x and y

$$\text{cost function} = J = \frac{1}{n} * \sum_{i=1}^n (y_{a_i} - y_{p_i})^2$$

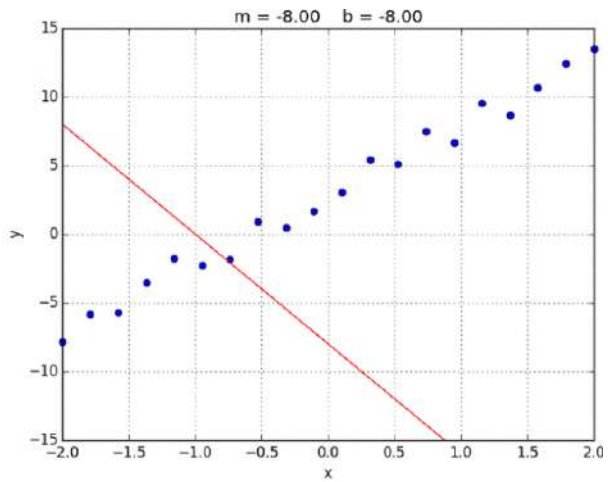
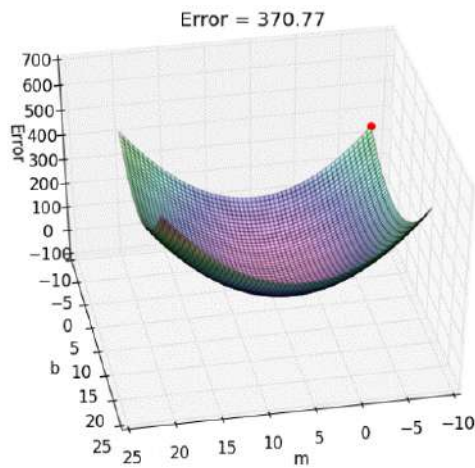
$$y_{p_i} = b_o + b_1 * x_1$$

$$\text{cost function} = J(b_o, b_1) = \frac{1}{n} * \sum_{i=1}^n (y_{a_i} - (b_o + b_1 * x_1))^2$$

$$\text{cost function} = J(b_o, b_1) = \frac{1}{n} * \sum_{i=1}^n (y_{a_i} - b_o - b_1 * x_1)^2$$

$$y_{a_i} = y^{\wedge} \text{ (hat means actual)}$$

Goal: Find the suitable coefficients in order to Minimize the Cost function



Goal: Find the suitable coefficients in order to Minimize the Cost function

Minimum point means slope = 0

slope means = differentiation = $\frac{dy}{dx}$

so we need to differentiate the cost function (j) and makes equal to zero

but cost function has two parameters (b_0, b_1)

so we need to partial differentiation

Case – 1: $\frac{\partial j}{\partial b_0} = 0$ then we wil get b_0

Case – 2: $\frac{\partial j}{\partial b_1} = 0$ then we wil get b_1

The entire procedure known as OLS(ordinaty least square)Method